

Lab 2: A price you can take to a house that has not sold yet

70-467 - Machine Learning for Business Analytics

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A listing desk in Ames, Iowa, prices houses that sold between 2006 and 2010. When a house comes on, the sheet has living area, overall quality, baths, garage, year built, kitchen, and neighborhood. Someone has to print an ask. If it is high, the house sits. If it is low, they leave dollars a buyer would have paid. They also have to explain why this house is priced that way.

You have 2,925 of those sales, in dollars. (Five sales with more than 4,000 square feet of living area are already out of the file.) Mean price is about \$180,000. After the deal closes they also record how it closed (Normal, Partial, family, foreclosure) and whether it was new construction. Partial and New sales in this file sit around \$275,000 on average; the rest sit around \$172,000. The codebook on the next pages gives type and unit for every column.

Staff do not use the sheet in one way. Size is the obvious start. In this file overall quality moves more, in dollars, than a chunk of extra square feet: one quality step is tens of thousands of dollars holding the other listing facts. Some people want a separate number for each of 28 neighborhoods, because the same spec does not sell for the same money on every street. NridgHt versus Edwards is about \$35,000 on one listing-style line. Some people would bump the ask when recent Partial or New sales came in high. Those labels are how the sale later closed. They are not on the flyer when you print it.

Last week it was enough to quote residual on the listings you already priced. This week the number that matters is how far off the rule is on sales you did not use to fit. A line that looks tight on all 2,925 can fail on a neighborhood that was barely in that half. Coefficients along the way are how you defend which listing facts move dollars, holding the other recorded features fixed.

Your assignment is to produce an asking price for a house that is not in this file, using only facts you would have before it closes, and to say how far off that rule was on sales it did not use to fit, next to the \$180,000 mean. Along the way you will have to decide what to do with the 28 neighborhood names and with Partial and New.

House A is already in the file. Use a house with about 1,440 square feet of living area, overall quality 6, two full baths, a two-car garage, built in the early 1970s, kitchen typical. You can check the rule against sales like that.

House B is not in the file. Same spec. One version of the desk would price it as if it sat in NridgHt; another as if it sat in Edwards. There is no listed price to look up.

The file does not contain days on market, the cost of sitting, or how many extra buyers would appear if the ask were cut by \$10,000. Estimates from this file are associations among Ames sales in 2006–2010. They do not measure the effect of changing a price, and they are not a 2026 valuation.

Read the codebook before class. Bring a laptop. In the room you will work in R, then post a number and the residual on sales you did not use to fit. After class you will have additional questions connected to the evaluation slides.

Codebook. 2,925 rows, 82 columns. One row = one residential sale in Ames, 2006–2010. Cite De Cock (2011), *Journal of Statistics Education*.

Variable	Type	Unit	Description
Order	numeric	none	Row index in the original file.
PID	numeric	none	Parcel ID. Not a predictor.
MS.SubClass	numeric	code	Building class (one-story, two-story, ...).
MS.Zoning	factor	none	Zoning: RL, RM, ...
Lot.Frontage	numeric	feet	Street frontage. Missing on 490 rows.
Lot.Area	numeric	sq ft	Lot size.
Street	factor	none	Paved or gravel.
Alley	factor	none	Alley access. Missing = no alley (most rows).
Lot.Shape	factor	none	Lot shape.
Land.Contour	factor	none	Flatness.
Utilities	factor	none	Utilities available.
Lot.Config	factor	none	Inside, corner, cul-de-sac, ...
Land.Slope	factor	none	Slope.
Neighborhood	factor	none	28 neighborhoods in Ames.
Condition.1	factor	none	Proximity to main road or railroad.
Condition.2	factor	none	Second proximity condition if any.
Bldg.Type	factor	none	One-family, townhouse, duplex, ...
House.Style	factor	none	One story, two story, ...
Overall.Qual	numeric	1–10	Overall material and finish. Large in dollars in this file.
Overall.Cond	numeric	1–10	Overall condition.
Year.Built	numeric	year	Year built.
Year.Remod.Add	numeric	year	Remodel year (same as built if none).
Roof.Style	factor	none	Roof style.
Roof.Matl	factor	none	Roof material.
Exterior.1st	factor	none	Exterior covering.
Exterior.2nd	factor	none	Second exterior if any.
Mas.Vnr.Type	factor	none	Masonry veneer type.
Mas.Vnr.Area	numeric	sq ft	Masonry veneer area.
Exter.Qual	factor	none	Exterior quality.
Exter.Cond	factor	none	Exterior condition.
Foundation	factor	none	Foundation type.
Bsmt.Qual	factor	none	Basement height/quality. Missing = no basement.
Bsmt.Cond	factor	none	Basement condition.
Bsmt.Exposure	factor	none	Walkout or garden exposure.
BsmtFin.Type.1	factor	none	Rating of finished basement area 1.
BsmtFin.SF.1	numeric	sq ft	Type 1 finished sq ft.
BsmtFin.Type.2	factor	none	Rating of finished basement area 2.
BsmtFin.SF.2	numeric	sq ft	Type 2 finished sq ft.
Bsmt.Unf.SF	numeric	sq ft	Unfinished basement sq ft.
Total.Bsmt.SF	numeric	sq ft	Total basement sq ft.
Heating	factor	none	Heating type.
Heating.QC	factor	none	Heating quality.
Central.Air	factor	none	Central air Y/N.
Electrical	factor	none	Electrical system.
X1st.Flr.SF	numeric	sq ft	First floor sq ft.
X2nd.Flr.SF	numeric	sq ft	Second floor sq ft.
Low.Qual.Fin.SF	numeric	sq ft	Low-quality finished sq ft.
Gr.Liv.Area	numeric	sq ft	Above-grade living area. Five sales over 4,000 sq ft already dropped.
Bsmt.Full.Bath	numeric	count	Basement full baths.
Bsmt.Half.Bath	numeric	count	Basement half baths.
Full.Bath	numeric	count	Full baths above grade.
Half.Bath	numeric	count	Half baths above grade.
Bedroom.AbvGr	numeric	count	Bedrooms above grade.
Kitchen.AbvGr	numeric	count	Kitchens above grade.
Kitchen.Qual	factor	none	Kitchen quality.
TotRms.AbvGrd	numeric	count	Rooms above grade (not baths).
Functional	factor	none	Home functionality.
Fireplaces	numeric	count	Number of fireplaces.
Fireplace.Qu	factor	none	Fireplace quality. Missing = no fireplace.
Garage.Type	factor	none	Garage location. Missing = no garage.
Garage.Yr.Blt	numeric	year	Year garage built.
Garage.Finish	factor	none	Interior finish of garage.
Garage.Cars	numeric	count	Car capacity. One missing row in the raw file.
Garage.Area	numeric	sq ft	Garage area.
Garage.Qual	factor	none	Garage quality.
Garage.Cond	factor	none	Garage condition.
Paved.Drive	factor	none	Paved driveway.
Wood.Deck.SF	numeric	sq ft	Wood deck.

Variable	Type	Unit	Description
Open.Porch.SF	numeric	sq ft	Open porch.
Enclosed.Porch	numeric	sq ft	Enclosed porch.
X3Ssn.Porch	numeric	sq ft	Three-season porch.
Screen.Porch	numeric	sq ft	Screen porch.
Pool.Area	numeric	sq ft	Pool area (almost all zero).
Pool.QC	factor	none	Pool quality. Missing on every remaining row.
Fence	factor	none	Fence quality. Missing = no fence.
Misc.Feature	factor	none	Shed, etc. Missing = none.
Misc.Val	numeric	USD	Value of miscellaneous feature.
Mo.Sold	numeric	month	Month the sale closed.
Yr.Sold	numeric	year	Year the sale closed (2006–2010).
Sale.Type	factor	none	How it sold (WD, New, COD, ...). Recorded at closing.
Sale.Condition	factor	none	Normal, Partial, Abnorml, Family, ...Recorded at closing. Partial $n = 242$.
SalePrice	numeric	USD	Sale price. This is Y . Mean about \$180,412 after the living-area screen.